Zhilei Chen
Guangdong Peizheng College,
Data and Computer Science,
53 Peizheng Rd., Chini Town,
Huadu, Guangzhou,
Guangdong 510830, China
2604513@peizheng.edu.cn

The Editor
International Journal of Fatigue
Elsevier

**Date:** 23 July 2026

**Re:** Submission of "Quantifying Method-Induced Uncertainty in Mechanical Reliability: A Factorial Variance-Decomposition Framework Demonstrated on Gear Bending Fatigue"

Dear Editor,

I am pleased to submit the above manuscript for consideration for publication in the *International Journal of Fatigue*.

**What this paper does.** When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes from methodological choices: which standard to use (ISO 6336, AGMA 2001, or FKM), how to correct for mean stress, what S–N curve to trust, how to penalize surface roughness. Each choice, studied in isolation by prior work, has been shown to matter. What has never been done—and what we present here—is to place all of these choices into a single controlled factorial experiment and decompose the total log-variance into its constituent sources, ranked by magnitude.

Using a full-factorial ANOVA over 144 combinations (5 switches, 116 finite-life entries), we find that two factors account for 62.0% of total variance: the choice of standard (31.1%) and the choice of mean-stress correction (30.9%). Surface roughness (8.5%) and a Standard $\times$ Rz interaction (5.0%) are secondary. The S–N curve source and cumulative damage rule—factors that the literature often treats as consequential—are negligible (<1% each). Bootstrap validation with 2000 draws confirms stable rankings. A multi-torque comparison reveals that the factor ranking is operating-point-dependent, not a universal constant.

All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.). S–N parameters are from the publicly accessible Waterloo SMDId-base. The analysis pipeline is released as an open-source, auditable toolkit.

**How this differs from prior work—including recent contributions in this journal.** We are aware that the *International Journal of Fatigue* has recently published important work on standard-to-standard divergence in gear fatigue assessment, notably

Bonaiti et al. (*Int. J. Fatigue*, Vol. 181, 2024, 108145), who compared ISO 6336 and AGMA 2001-D04 predictions against pulsator test data. That study provides an invaluable experimental benchmark. Our manuscript complements and extends it in three ways:

1. **From binary comparison to factorial decomposition.** Bonaiti et al. ask whether ISO and AGMA differ (yes, they do). We ask: across all major methodological switches an engineer faces, how is the total log-variance partitioned—and which switch contributes the most? The factorial framework reveals that mean-stress correction (not examined by Bonaiti et al.) is statistically indistinguishable from standard choice as the largest variance source.

2. **From two dimensions to five.** Bonaiti et al. examine standard choice and statistical elaboration method. We simultaneously vary five independent choices (standard, mean-stress correction, roughness grade, S–N source, damage rule), capturing interactions that binary comparisons miss.

3. **From qualitative conclusion to quantitative ranking.** "Standards differ" is a qualitative statement. "The standard accounts for 31.1% of variance with a bootstrap S/N ratio of 5, while mean-stress correction accounts for 30.9% with S/N of 4" is a quantitative one that enables engineering resource allocation: which uncertainty sources warrant the most modeling effort?

4. **Bootstrap noise floor.** No prior gear fatigue UQ study has employed bootstrap resampling to distinguish genuine signal from finite-sample fluctuation, making our uncertainty budget a repeatable measurement rather than a one-shot estimate.

The two studies are complementary, not competing: Bonaiti et al. provide independent experimental evidence that the standard-choice effect exists; we measure its magnitude relative to four other uncertainty sources and provide a reusable framework applicable to any mechanical reliability calculation where multiple handbooks or methods compete.

**Fit with the journal.** The *International Journal of Fatigue* is the natural home for this work. The journal has published gear bending fatigue studies (Glodez et al., 2002; Bonaiti et al., 2024), cumulative damage surveys (Fatemi & Yang, 1998), and mean-stress methodology papers—each examining one factor in isolation. Our manuscript synthesizes these threads within a unified UQ framework, and we explicitly cite and build upon the journal's own published record. The methodology is not limited to gears: it applies to any mechanical component where multiple standards or correction methods compete.

**Additional information.** This manuscript has not been published previously and is not under consideration elsewhere. All authors have approved the submission. The authors declare no competing financial interests.

**Use of AI-assisted tools.** During the preparation of this work, the author used Claude (Anthropic) as a language and analysis assistant for literature search, code verification against ISO/AGMA/FKM standards, and manuscript drafting. All numerical results were generated by the open-source analysis pipeline included with this submission and verified by the `paper_check.py` audit tool. The author takes full responsibility for the integrity of all data, citations, and conclusions presented herein.

Sincerely,

Zhilei Chen